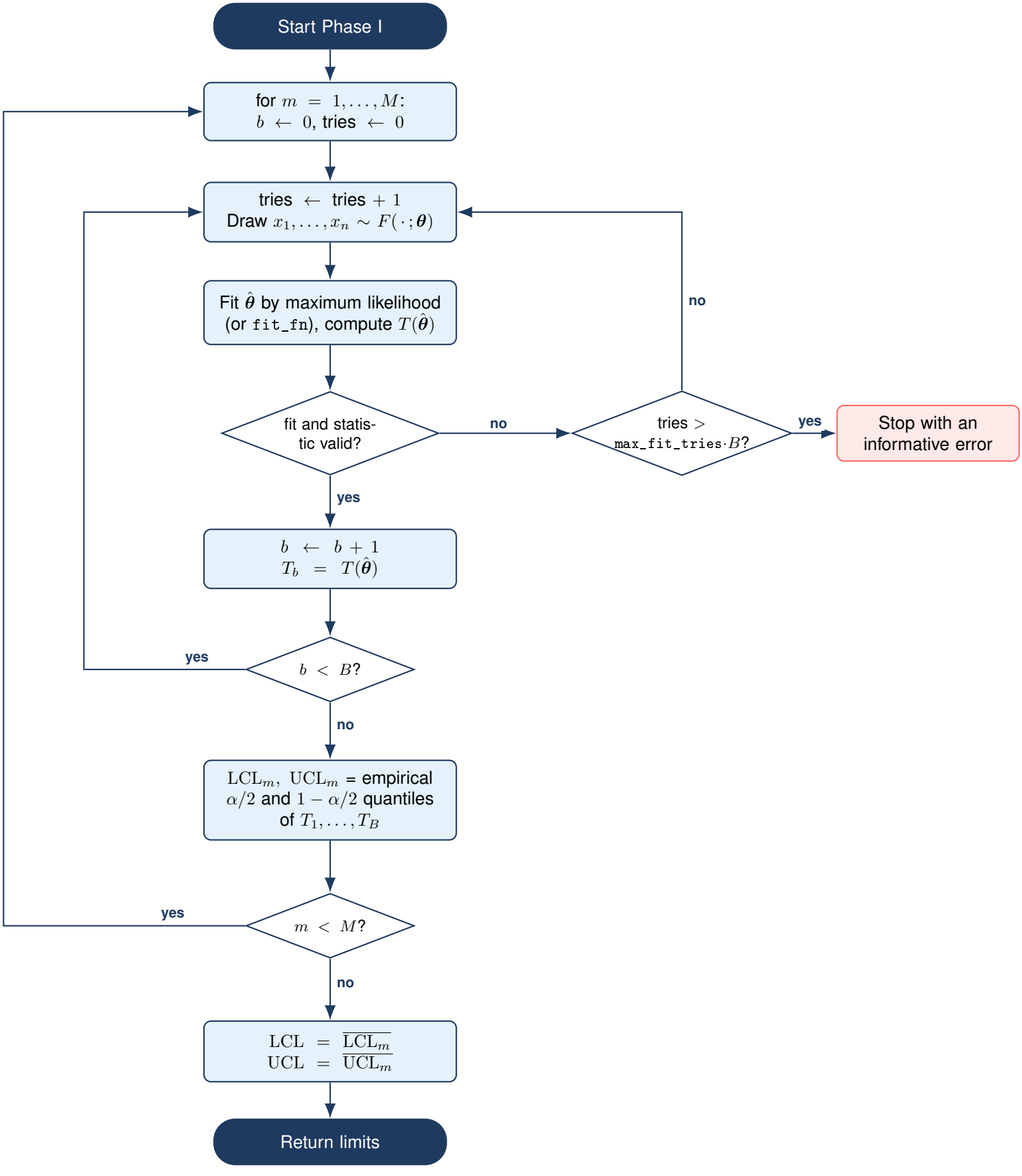

Start Phase I
for $m = 1, \ldots, M$:
$b \leftarrow 0$, tries $\leftarrow 0$
tries $\leftarrow$ tries $+ 1$
Draw $x_1, \ldots, x_n \sim F(\cdot\,; \boldsymbol{\theta})$
Fit $\hat{\boldsymbol{\theta}}$ by maximum likelihood (or fit_fn), compute $T(\hat{\boldsymbol{\theta}})$
fit and statistic valid?
no
tries > max_fit_tries·B?
yes
Stop with an informative error
no
yes
$b \leftarrow b + 1$
$T_b = T(\hat{\boldsymbol{\theta}})$
$b < B$?
yes
no
LCL_m, UCL_m = empirical $\alpha/2$ and $1 - \alpha/2$ quantiles of $T_1, \ldots, T_B$
$m < M$?
yes
no
$\mathrm{LCL} = \overline{\mathrm{LCL}_m}$
$\mathrm{UCL} = \overline{\mathrm{UCL}_m}$
Return limits